

Cerberus - How Radix achieves infinite linear scalability while preserving atomic composability

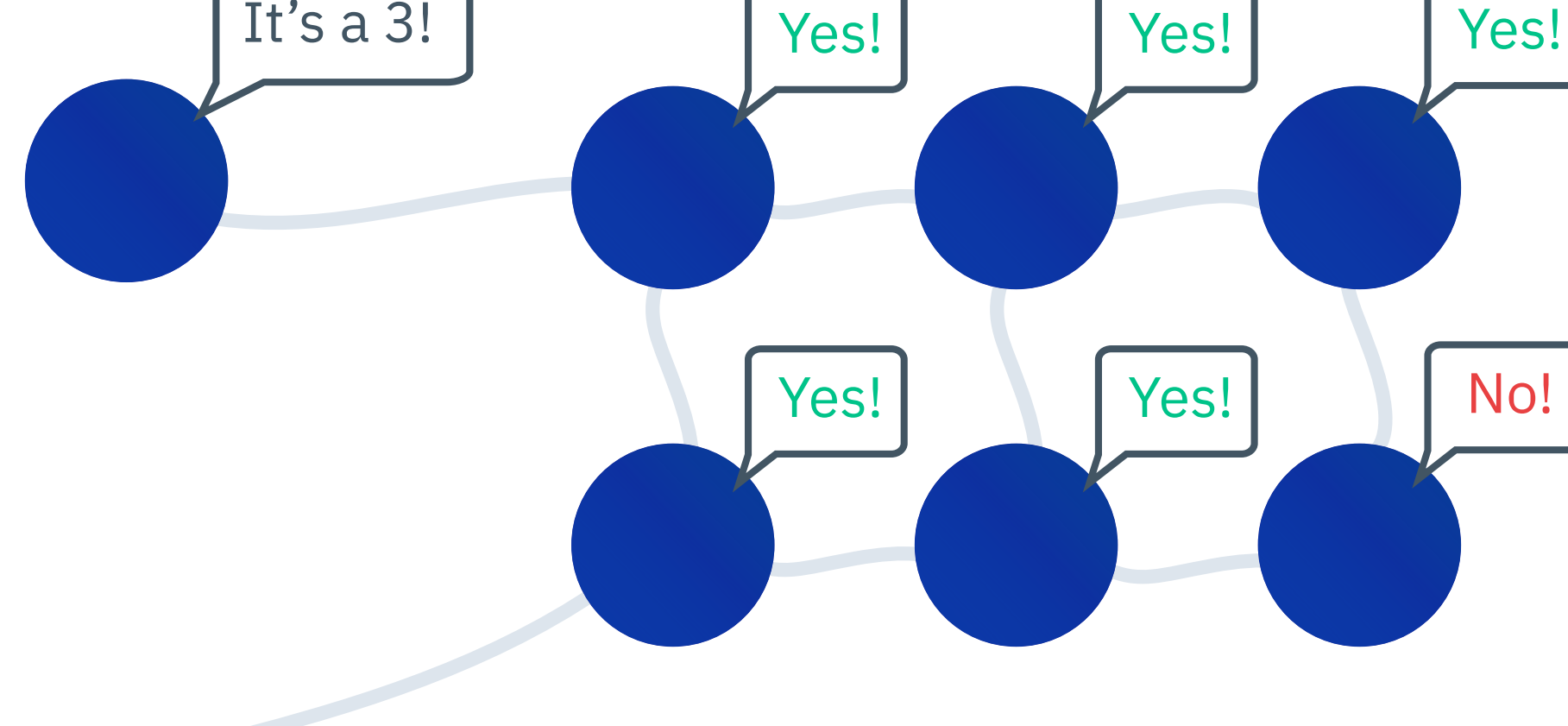
Nakamoto vs BFT-style Consensus

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So before jumping into Cerberus, we go back a step and first provide a little background on consensus.

Consensus is a problem of coordination between nodes - getting them to either agree or reject something, together.

This means the nodes need a protocol - some established set of procedures and rules, that governs how they communicate with one another and come to a decision.



Nakamoto Consensus

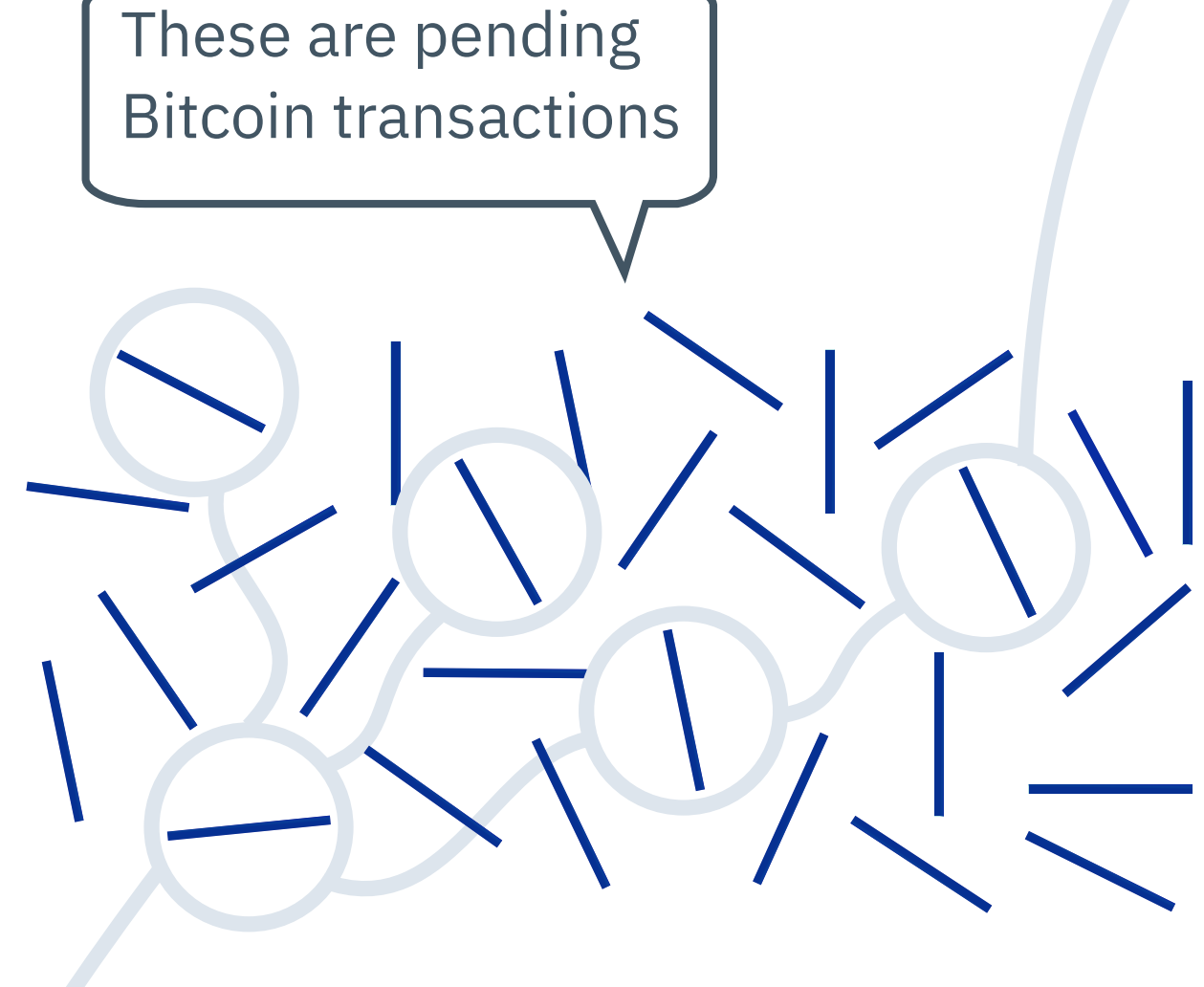
Bitcoin uses a consensus protocol some people call "Nakamoto" consensus.

In Bitcoin, nodes select transactions that are pending from users, and attempt to solve a cryptographic puzzle - performing lots of computation to find a special kind of hash derived from the data of those selected transactions.

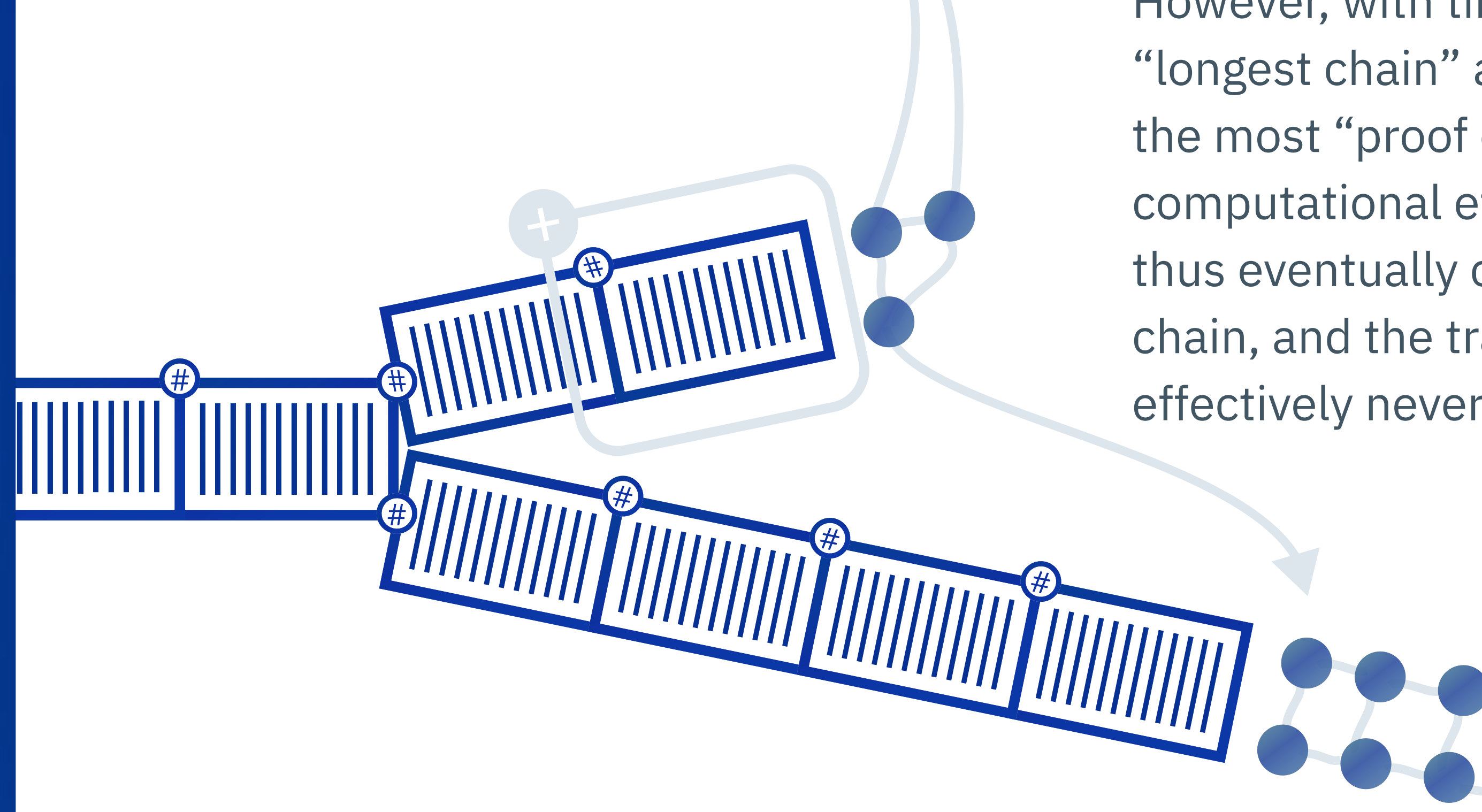
The first node to find that hash then writes a block with those selected transactions and proposes this to the network. If other nodes agree that the block and hash are valid, then those nodes also write that block to their own copy of the blockchain, and that block propagates throughout the network.

If nodes don't agree, we have what's called a "fork" such as that depicted to the bottom left. This is where some nodes maintain a record of one set of blocks, and other nodes keep a record of another set.

However, with time, nodes will migrate over to the "longest chain" as that is the chain that can demonstrate the most "proof of work" - the chain that had the most computational effort that went into producing it. Nodes thus eventually come to consensus around the longest chain, and the transactions in the shorter chain will effectively never have happened.



Bitcoin "Mempool"



Where there is a fork, nodes will eventually migrate and come to consensus around the "longest chain".

Because of this, Nakamoto consensus is considered "probabilistic", as the node that proposes the transactions that go into a particular block is not predetermined; and you can never be 100% sure when a transaction is "final", as there might always be a longer chain out there.

However, over time, as a chain gets longer and longer, the chances of there being another chain with more proof of work effectively becomes so small, that you can be almost certain that your transaction is on the longest chain, making it effectively final. This is why some exchanges require you to wait 60 minutes, even though the Bitcoin block time is roughly 10 minutes.

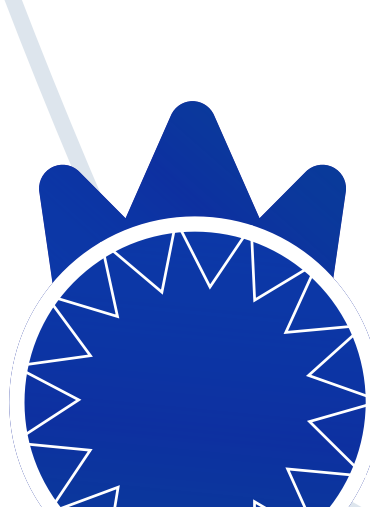
BFT-style Consensus

Cerberus falls under a different class of consensus protocol - one that can be referred to as "Byzantine Fault Tolerant style", and can be considered to be "deterministic".

This is because this style of consensus has clear rules about who can vote on a transaction and exactly how many votes are needed before everyone can agree the

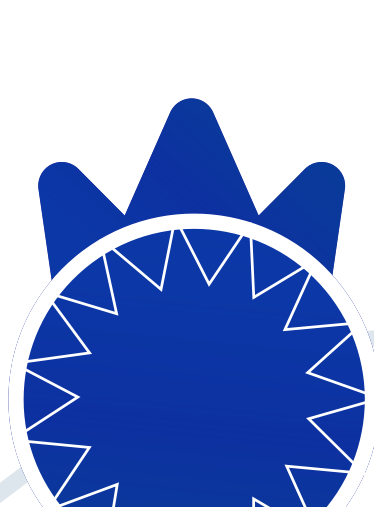
transaction is 100% final. Once a transaction is finalized, there is no way of there being a "longer chain" out there, as there is no uncertainty in the process.

In this class of consensus algorithm, the node that gets to propose the next transaction is called the "leader" of that particular consensus round.



This is a leader.

The role of leader is to...



...submit messages to the nodes within their validator set and to drive progress through "phases"...

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...collect the votes from nodes on whether they agree with the message...

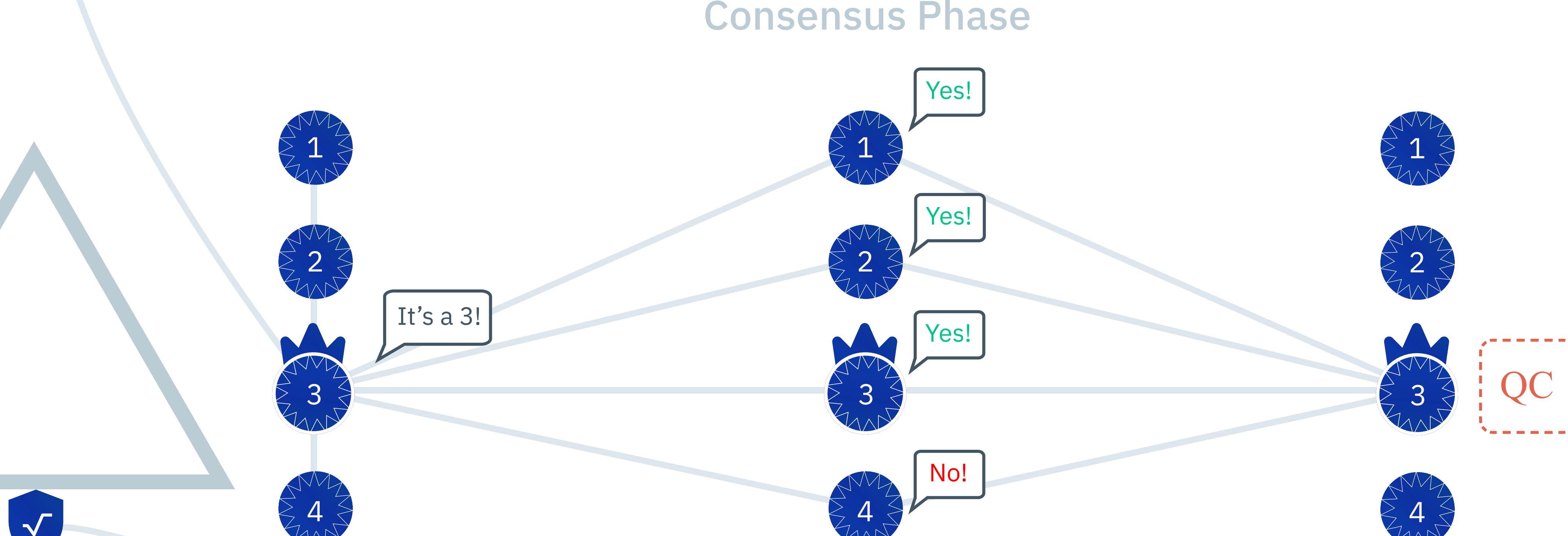


...and to then broadcast the aggregation of those votes back to the nodes, proving that enough nodes agreed, in what's called a Quorum Certificate.

Phase

A phase is a series of steps that a collection of nodes undertake within consensus. Here is what a typical consensus phase looks like for a hypothetical validator set with just four nodes (in reality there would be many more). A phase is composed of three steps:

Consensus Phase



1. Broadcast

Here we can see the leader broadcasting a message to the other nodes in the validator set.

The message in the first phase is a proposal. In later phases, the message is the QC from the previous phase (we explain what this means later).

2. Vote

All nodes then vote on whether the message is valid. They do this by cryptographically signing their vote, and sending their vote to the leader.

3. Create Quorum Certificate

The leader then gathers the votes and creates a Quorum Certificate - cryptographic proof that enough votes were in agreement with the message.

If there aren't enough votes, no valid QC can be produced.

Choosing a leader

In Cerberus, the leader within each validator set shuffles periodically, after every "consensus round".

The selection process is deterministic, and is agreed between nodes in a similar way to the consensus phase described above. Nodes therefore know, at any given point in time, who the leader of that validator set is. In consensus lingo, when a leader changes, this is called a "view change".

So those are the basics. Let's now move on to local Cerberus.

The version of Cerberus described in this infographic series is scheduled to launch as part of the fully sharded Radix Xi'an release. Please visit www.radixdlt.com for details on the Radix roadmap.